*DMET 909 • VIDEO GAME DESIGN*

# FINAL APPROACH

Assignment 02 — Game World, Mechanics & Prototype

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Submission date: 16 Oct 2026

Course: DMET 909 — Video Game Design

Project version: Assignment 02 prototype

## Submission links

| Deliverable | Editable link placeholder |
|---|---|
| GitHub repository | https://github.com/Darsh2984/FinalApproach-VGD-A01- |
| Gameplay demonstration (Google Drive) | https://drive.google.com/file/d/1SmopwkuLNCGuKtm9kaKBW1Skbz-q0nOG/view |

# 1. Executive overview and concept

## 1.1 Working title

Final Approach

## 1.2 Concept statement

Final Approach is a single-player, physics-based arcade javelin game in which rhythmic run-up input, launch angle, release position, and stability combine to determine a measurable throw. The player prepares in an interactive locker room before entering a stadium for a three-attempt precision challenge.

## 1.3 Genre and differentiators

Primary genre: arcade sports / precision sports. Secondary design characteristics: timing, physics simulation, short-session score chasing, environmental interaction. The distinctive feature is the transparent relationship between preparation, speed, release timing, and projectile flight rather than a single-button distance roll.

## 1.4 Audience and platform

Intended audience: approximately ages 13–35, especially players interested in athletics, score chasing, rhythm-based input and rapid replay. This audience and the PC target are inherited from the earlier Assignment 01 concept; they are design intentions rather than measured demographic findings. Current development target: Windows PC using Unity 6.3 LTS and Universal Render Pipeline (URP). Hardware minimums have not been benchmarked and should not be invented.

## 1.5 Development status

| Area | Status | Evidence / caveat |
|---|---|---|
| Locker room exploration | Implemented | Starter Assets third-person character and camera |
| Stadium javelin loop | Implemented | Run-up, aim, release, projectile flight, landing and score |
| Scene transition | Implemented | E interaction loads stadium scene |
| Cinemachine and lighting | Implemented | Three behaviors each, described in §6 |

## 1.6 Scope and monetization

This is a course prototype, not a released commercial game. No monetization, microtransactions or online economy are implemented. The intended experience is a compact single-player session, with replayability based on improving release quality and distance.

# 2. Player experience, objectives and progression

## 2.1 Player role and emotional arc

The player assumes the role of a javelin athlete. The locker room establishes anticipation and physical preparation; crossing into the stadium marks the transition from exploration to focused competition. During the run-up, rhythmic input produces urgency; the release requires precision; the flight camera creates suspense; the landing result provides immediate feedback and encourages another attempt.

### 2.2 Core gameplay loop

Explore preparation room → optionally open locker / collect equipment / replenish stamina / change lighting → activate stadium entrance → accelerate using Space taps → hold Left Shift to prepare → adjust angle with W/S → release Left Shift → observe Cinemachine javelin tracking → read result → press Enter for next attempt (up to three attempts).

### 2.3 Objectives and outcomes

| Objective | Mechanic / feedback |
|---|---|
| Primary | Achieve the greatest legal javelin distance |
| Precision | Release close to the foul line without crossing it |
| Execution | Build run-up speed while managing stability |
| Session progression | Complete a three-attempt sequence and compare results |
| Failure state | Crossing the foul line produces FOUL and zero distance |
| Timeout | Countdown reaches zero |

### 2.4 Key moments

Moment A — entering the stadium: player-initiated transition from preparation to competition.
Moment B — release: angle, speed, and distance from the foul line converge into a single high-stakes input.
Moment C — flight and landing: Cinemachine switches to the projectile and the measured result closes the feedback loop.

### 2.5 Game duration and progression limits

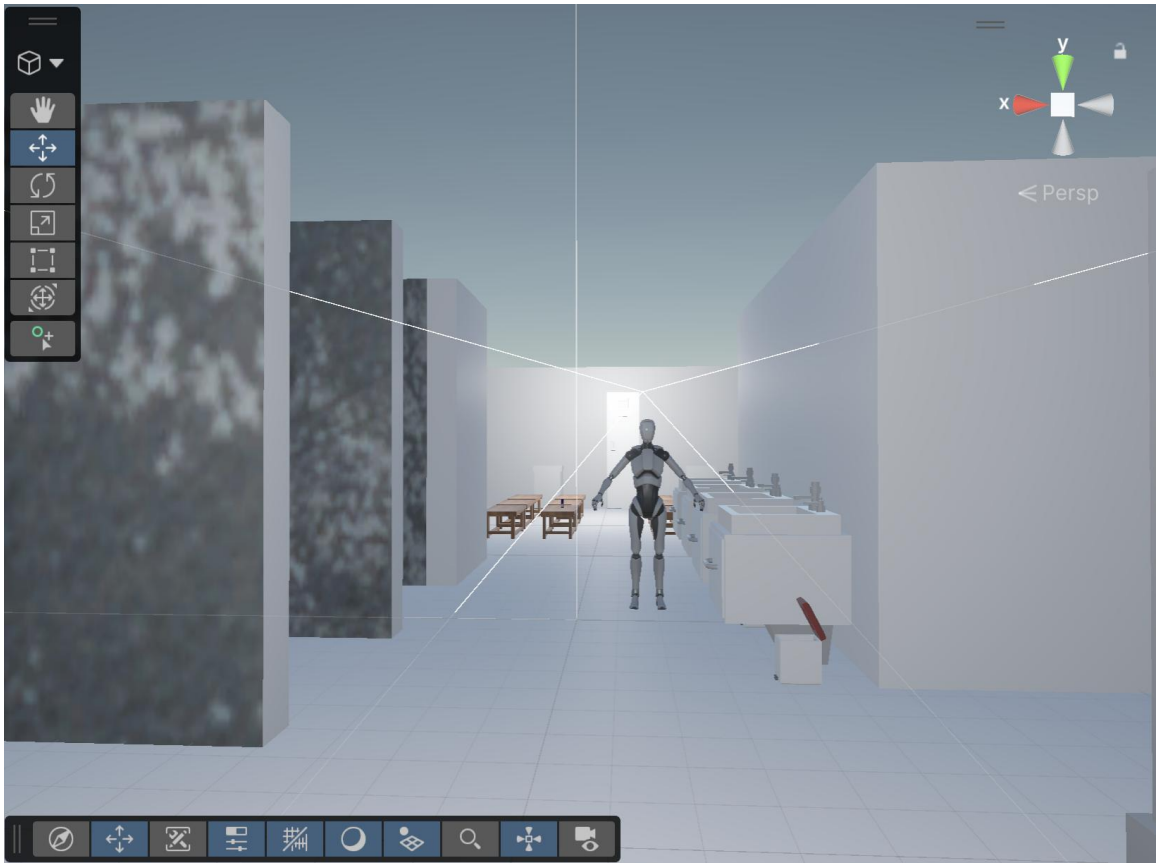
A run comprises three throwing attempts. Actual session duration varies with locker-room exploration and player input; it has not been measured. No leveling, skill trees, currency progression or NPC questline is implemented.

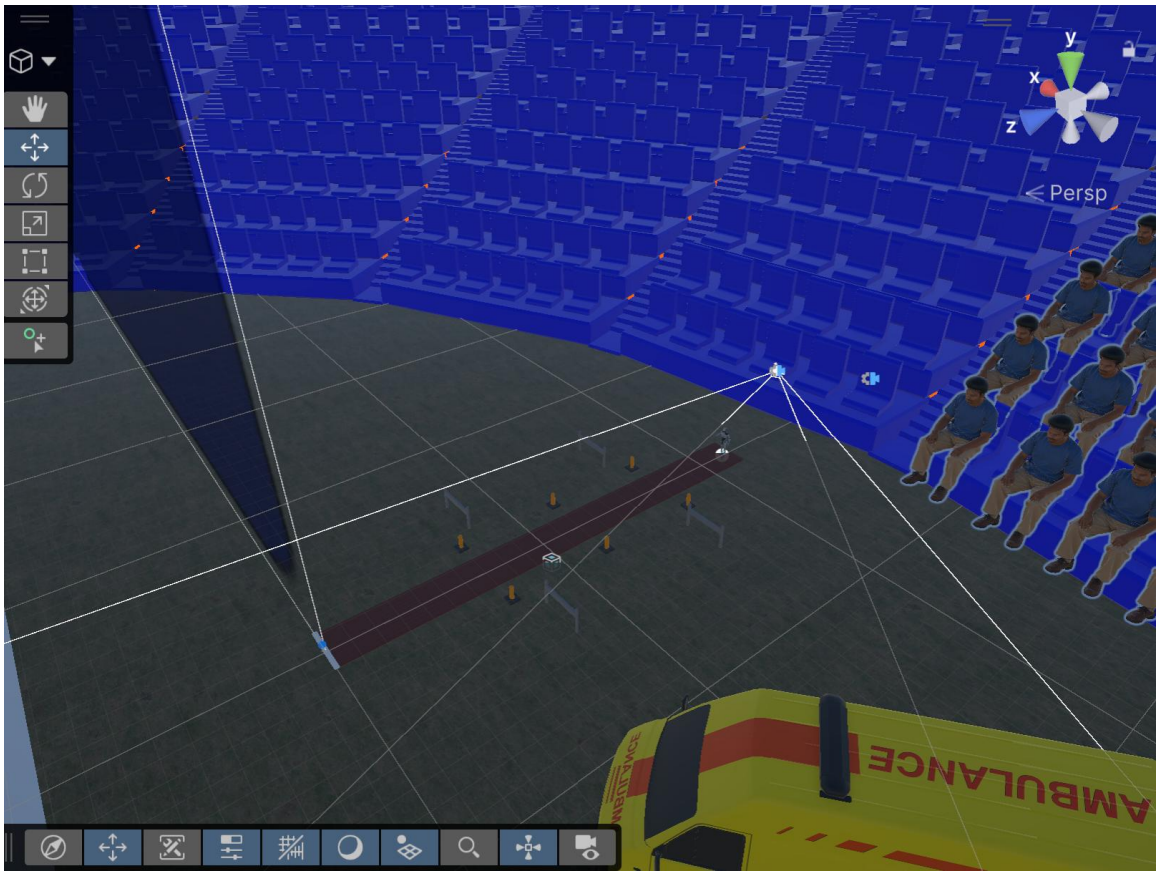
## 3. Game world

### 3.1 World organization and locations

| Location | Characteristics | Player affordances |
|---|---|---|
| Athlete Preparation / locker room | Interior, enclosed walls, solid collision floor, locker, equipment rack, energy drink station, lighting and entrance | Walk, inspect, interact, restore stamina, transition to stadium |
| FinalApproach Stadium | Outdoor-style athletics setting with throwing approach, foul line, landing field, javelin physics and scoring HUD | Build speed, choose angle, throw, watch landing, retry |

The two scenes are discrete spaces connected by a deliberate interaction, not a physically continuous open world. The stadium entrance presents “[E] Enter Stadium” and calls Unity SceneManager.LoadScene("FinalApproach_Stadium"). The transition is one-way in the currently confirmed prototype


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## 3.2 Spatial topology and navigation

The locker room supports free third-person walking within collider-bounded geometry. The stadium uses a constrained javelin approach along the forward Z direction, followed by a simulated projectile travelling into the landing field. The distinct control schemes intentionally separate preparation/exploration from competitive execution.

## 3.3 Physical scale and proportionality

Unity world coordinates are treated as meters for documentation and physics design. Exact dimensions of the imported locker room, runway and landing field have NOT been measured from the final Unity scene.

## 3.4 Boundaries and restrictions

Locker room: walls and an explicit Box Collider floor prevent the third-person CharacterController from falling through imported geometry. Stadium: the foul line is a gameplay boundary at throwLineZ = 0 in the supplied ThrowController configuration; a release beyond this line is a foul. The landing-field Collider defines valid landing evaluation. Verify stadium perimeter barriers and whether players can leave the runway.

## 3.5 Temporal dimension

The game advances in real time using Unity Time.deltaTime for locomotion, smoothing, light behavior and timer display. A 10-second PreparationTimer countdown is implemented as a visual timer, beginning at 10 and turning red in the last three seconds. Integration that forfeits a timed-out attempt and advances the three-attempt game state is NOT confirmed. The energy drink reappears after 15 seconds, and stamina decreases over time while moving. These time-dependent systems give the world a temporal dimension without a day/night cycle.

### 3.6 Environment as mechanics

The environment is not only decorative: locker doors rotate in response to E, the rack javelin disappears when collected, the energy drink restores stamina and respawns, the light switch toggles two lights, entrance proximity controls a light, and the entrance loads the competition scene.

## 4. World-object inventory and verification

Assignment 02 requests at least 30 different objects with role, significance and usability. The following is a 30-row audit worksheet. “Confirmed” means the object/system appeared in the development walkthrough; “Verify” means the precise instance or independent identity still needs checking in the Unity Hierarchy. This table must not be used to claim 30 verified distinct objects before completing the audit.

| # | Object / distinct role | Location / usability | Status |
|---|---|---|---|
| 1 | PlayerArmature | Locker room / stadium — Controllable athlete visual | Confirmed |
| 2 | Player gameplay root | Stadium — Run and throw controllers | Confirmed |
| 3 | Locker | Locker room — Interactive storage fixture | Confirmed |
| 4 | Locker door | Locker room — Rotates open/closed | Confirmed |
| 5 | DoorPivot | Locker room — Door hinge transform | Confirmed |
| 6 | Javelin rack | Locker room — Equipment display | Confirmed |
| 7 | Equipment javelin | Locker room — Collectible visual disappears | Confirmed |
| 8 | Energy drink can | Locker room — Consumable stamina refill | Confirmed |
| 9 | EnergyDrinkStation | Locker room — Respawn and interaction logic | Confirmed |
| 10 | Light switch | Locker room — Manual environmental control | Confirmed |
| 11 | LockerRoomLight A | Locker room — Toggleable point light | Confirmed |
| 12 | LockerRoomLight B | Locker room — Toggleable point light | Confirmed |
| 13 | StadiumEntrance | Locker room — Scene transfer trigger | Confirmed |
| 14 | EntranceProximityLight | Locker room — Automatic distance light | Confirmed |
| 15 | LockerRoomFloorCollider | Locker room — Collision support | Confirmed |
| 16 | InteractionCanvas | Locker room — Context UI container | Confirmed |
| 17 | EnergyBar | Locker room — Stamina feedback | Confirmed |
| 18 | Locker prompt | Locker room — Open/close interaction cue | Confirmed |
| 19 | Javelin prompt | Locker room — Equipment pickup cue | Confirmed |
| 20 | Energy drink prompt | Locker room — Consumable pickup cue | Confirmed |

| | | | |
|---|---|---|---|
| 21 | Stadium player camera | Stadium — Cinemachine follow | Confirmed |
| 22 | Stadium javelin camera | Stadium — Projectile tracking | Confirmed |
| 23 | Locker close-up camera | Locker room — Interaction cinematic | Confirmed |
| 24 | PlayerFollowCamera | Locker room — Third-person exploration | Confirmed |
| 25 | Javelin projectile prefab | Stadium — Launched Rigidbody object | Confirmed |
| 26 | Landing field collider | Stadium — Landing validation | Confirmed |
| 27 | Foul line marker | Stadium — Throw validity threshold | Confirmed |
| 28 | Stadium spotlight | Stadium — Event-driven illumination | Confirmed |
| 29 | PreparationTimerText | Stadium — 10-second time display | Confirmed |
| 30 | Scoreboard / distance HUD | Stadium — Attempt result feedback | Confirmed |

## 5. Game mechanics catalogue

Assignment 02 requires 20 distinct mechanics. The catalogue below separates verified gameplay rules from supporting behaviors; it is NOT a claim that the rubric's 20 distinct mechanics threshold has been independently met. Related controls should not be double-counted without a defensible rule distinction.

| # | Mechanic | Specific rule / action | Status |
|---|---|---|---|
| 1 | Third-person exploration | WASD movement with CharacterController | Implemented |
| 2 | Interaction distance gating | E prompt only within proximity | Implemented |
| 3 | Locker open / close | E toggles hinged door rotation | Implemented |
| 4 | Equipment collection | E removes rack javelin | Implemented |
| 5 | Stamina depletion | Movement drains 3 units per second | Implemented |
| 6 | Stamina restoration | Energy drink refills to 100% | Implemented |
| 7 | Consumable respawn | Drink reappears after 15 seconds | Implemented |
| 8 | Manual light control | E toggles two point lights | Implemented |
| 9 | Automatic proximity light | Entrance light switches within 5 meters | Implemented |
| 10 | Scene transition | E loads stadium scene | Implemented |
| 11 | Rhythmic run-up | Repeated Space input drives forward approach | Implemented |
| 12 | Throw preparation | Hold Left Shift enters preparation state | Implemented |
| 13 | Angle adjustment | W/S adjusts throw angle 15°–60° | Implemented |
| 14 | Release timing | Shift release initiates throw | Implemented |
| 15 | Foul detection | Crossing Z=0 invalidates throw | Implemented |

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|---|---|---|---|
| 16 | Release-quality tiers | EARLY / SAFE / GOOD / PERFECT | Implemented |
| 17 | Run-speed contribution | CurrentSpeed influences launch velocity | Implemented |
| 18 | Stability modifier | High speeds reduce multiplier | Implemented |
| 19 | Projectile ballistics | Launch velocity and Rigidbody physics | Implemented |
| 20 | Landing evaluation | Collision/event computes distance | Implemented |
| 21 | Three-attempt flow | Enter begins next attempt | Implemented |
| 22 | Camera priority transitions | Cinemachine switches based on throw event | Implemented |
| 23 | Event spotlight | Launch on, landing off | Implemented |
| 24 | Preparation countdown | 10-second UI countdown | Implemented |

## 5.1 Movement and navigation

Locker room: Starter Assets ThirdPersonController moves a CharacterController and drives third-person animation. Stadium: PlayerRunController moves the gameplay root through a rhythmic Space-tap run-up; StadiumCharacterAnimator maps root displacement to Animator parameters Speed and MotionSpeed. Root motion remains disabled to prevent double movement. Jumping, swimming, climbing and vehicle traversal are not implemented and are not needed for the javelin sport concept.

## 5.2 Input and interaction constraints

E interactions are proximity-dependent. Current implementation uses independent scripts listening for E, so overlapping interaction zones could trigger more than one action; stations should be spatially separated or input centralized in a future revision. The javelin pickup removes the rack model but a held hand-javelin visual is not yet confirmed.

## 5.3 Javelin weapon/tool semantics

The javelin is sporting equipment, not a combat weapon. It has no damage, ammo, reload, NPC targets or combat range. Its gameplay effectiveness is measured by legal landing distance.

## 5.4 Verified numerical rules from ThrowController

| Parameter | Value / formula |
|---|---|
| Base throw velocity | 18 m/s (serialized default) |
| Run contribution | 0.9 × CurrentSpeed |
| Throw angle | 35° initial; clamp 15°–60° |
| Angle adjustment speed | 40°/second |
| Foul threshold | throwLineZ = 0 (serialized default) |
| Release-quality multiplier | distance > 3m: 0.70; >1.5m: 0.85; >0.5m: 0.95; otherwise 1.00 |
| Stability multiplier | speed ≤8: 1.00; ≤9: 0.97; >9: 0.93 |
| Launch speed | (18 + 0.9 × CurrentSpeed) × releaseQuality × stability |
| Forward / upward velocity | v cos(angle), v sin(angle) |
| Attempts | 3 per session (existing gameplay) |
| Preparation timer | 10 seconds |

# 6. Prototype evidence and behavior

## 6.1 Five world-changing player actions

| Action | Player input | Visible cause → effect |
|---|---|---|
| Open/close locker | E near locker | DoorPivot rotates; prompt updates |
| Collect javelin | E near rack | Equipment javelin disappears |
| Drink energy beverage | E near station | Energy bar fills; can disappears and respawns |
| Toggle room lights | E near switch | Two point lights switch state |
| Throw javelin | Hold/release Shift | Projectile spawns, travels and lands; result updates |
| Enter stadium (additional) | E near exit | Scene changes from locker room to stadium |

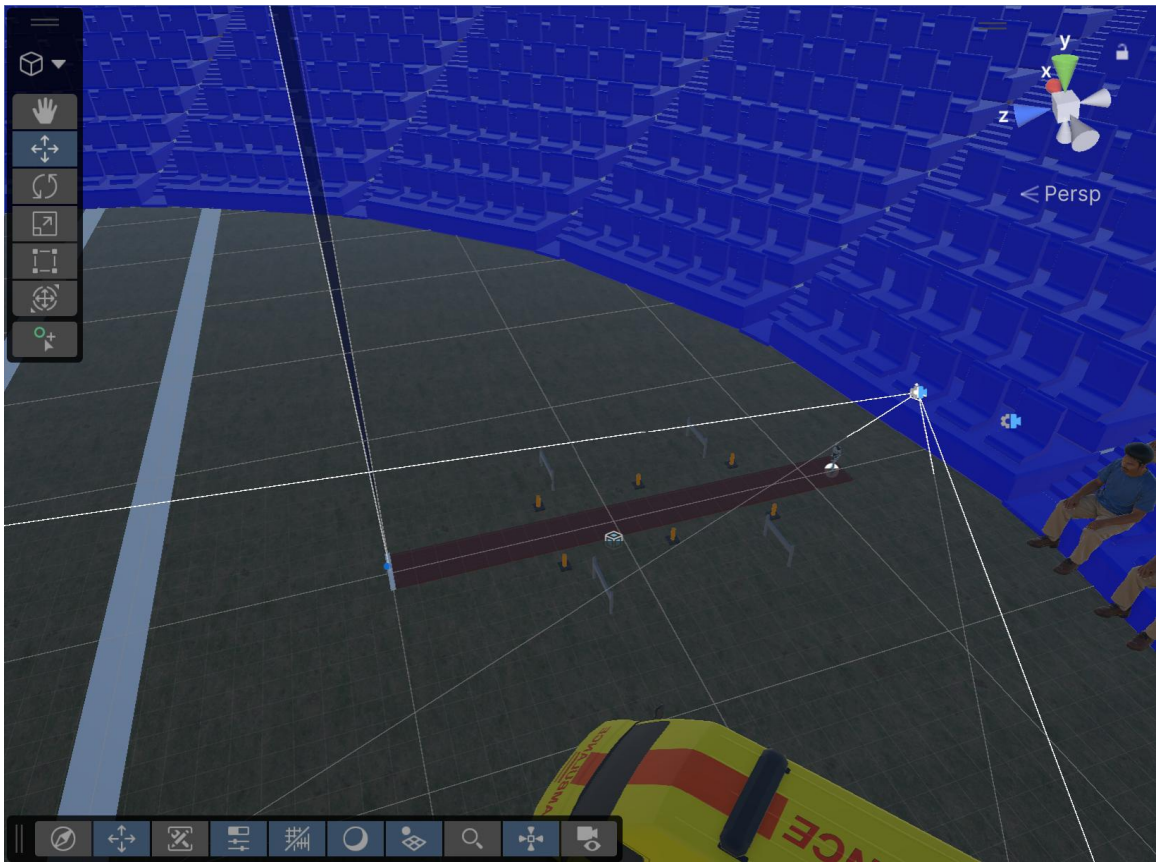
## 6.2 Three Cinemachine camera behaviors

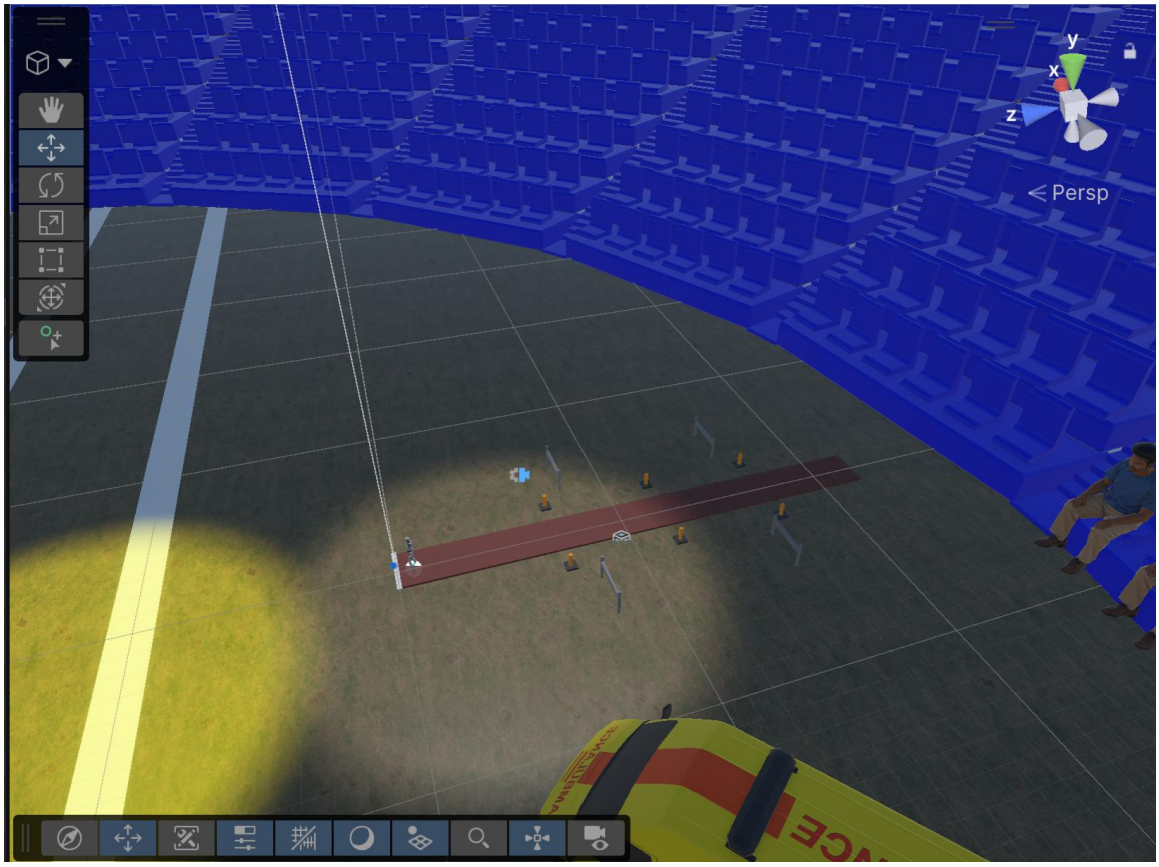
| Camera | Trigger / tracking | Implementation |
|---|---|---|
| Preparation follow | Third-person locomotion | PlayerFollowCamera tracks PlayerCameraRoot |
| Locker close-up | E locker interaction | LockerInteractionCamera priority temporarily rises, then returns |
| Stadium javelin flight | OnJavelinLaunched | StadiumJavelinCamera tracks spawned javelin; priority swap via CameraDirector |

StadiumPlayerCamera also provides pre-throw framing. The Main Camera uses Cinemachine Brain; its Default Blend duration is adjustable, with a tested short transition. The previous direct-transform CameraDirector was replaced by a priority/target-switching version.

## 6.3 Three distinct lighting behaviors

| Lighting behavior | Trigger | Effect |
|---|---|---|
| Manual room lighting | Player presses E on switch | Two locker-room point lights toggle |
| Proximity entrance lighting | Player within 5m of entrance | EntranceProximityLight enables; disables on exit |
| Event-driven stadium spotlight | ThrowController launch / landed events | Spotlight enables during flight, disables on landing |


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### 6.4 Demonstration video shot list

Recommended sequence: locker-room establishing shot → third-person walk → locker opening and close-up → javelin pickup → stamina drink and respawn → manual light switch → entrance proximity light → scene transition → run-up and angle → throw and Cinemachine follow → stadium spotlight → landing result → next attempt → timer display.

## 7. Interface, art, sound and accessibility

### 7.1 Controls

| Context | Input | Action |
|---|---|---|
| Preparation | WASD / mouse | Third-person movement / camera |
| Preparation | E | Interact with nearby station / door / entrance |
| Stadium | Space repeated | Rhythmic run-up |
| Stadium | Left Shift hold/release | Prepare and launch javelin |
| Stadium | W / S during preparation | Increase / decrease throw angle |
| Stadium | Enter | Next attempt |
| Stadium | H | How-to-play help |

### 7.2 HUD and interaction feedback

Preparation HUD: context-sensitive TMP interaction prompts and stamina slider. Stadium HUD: score, attempt/distance feedback, release quality, help overlay, and TIME: 10 countdown text. The countdown turns red in its last three seconds. Capture each screen and label key interface regions in the final report.

### 7.3 Visual and audio direction

Visual tone: readable sports venue with grounded environmental props, a recognizable third-person athlete, clear throwing field, and lighting that communicates interaction and event state. The character comes from Unity Starter Assets. The locker room and energy drink include imported 3D assets, with materials adjusted for URP. Stadium cheer/audio feedback was present in the earlier prototype. An exact audio asset inventory, license information and soundtrack plan require project inspection.

### 7.4 Help and usability

In-stadium H opens the existing how-to-play interface. Proximity prompts use the same E syntax to reduce cognitive load. Different movement schemes between locker room and stadium are deliberate but should be explained in the opening UI or demonstration. Color-only time warnings should ideally be paired with text or sound for accessibility.

## 8. Technical architecture and production

### 8.1 Engine and project organization

Engine: Unity 6.3 LTS, Universal Render Pipeline, C#. Core scenes: Assets/FinalApproach/Scenes/AthletePreparation.unity and Assets/FinalApproach/Scenes/FinalApproach_Stadium.unity. Unity Scene List includes both scenes. Imported character uses Starter Assets Third Person Controller; cinematic transitions use Unity.Cinemachine; text uses TextMeshPro.

## 8.2 Principal scripts

| Script | Responsibility |
|---|---|
| PlayerRunController | Stadium run-up movement and CurrentSpeed |
| ThrowController | Aim, release, foul detection, launch, landing event and reset |
| JavelinController | Projectile physics and landing callback |
| CameraDirector | Cinemachine tracking targets and priorities |
| StadiumCharacterAnimator | Translate stadium displacement into Animator speed |
| LockerInteraction | Door state, prompt and close-up camera |
| JavelinPickup | Equipment collection |
| EnergyDrinkPickup / PlayerStamina | Consumable refill, respawn and stamina drain |
| LightSwitchInteraction / ProximityLight | Manual and automatic lighting |
| JavelinSpotlightController | Event-driven stadium lighting |
| StadiumEntrance | SceneManager scene transition |
| PreparationTimer | 10-second countdown UI |

## 8.3 Event and data flow

ThrowController emits OnJavelinLaunched(Transform) → CameraDirector assigns javelin tracking target and raises camera priority; JavelinSpotlightController enables spotlight. JavelinController landing callback → ThrowController emits OnJavelinLanded → CameraDirector retains landing focus and spotlight switches off; OnThrowCompleted(bool, float, string) communicates foul/legal result and release quality to scoring.

## 8.4 Known technical risks

| Risk | Consequence | Mitigation |
|---|---|---|
| Independent E listeners | Two interactions could fire together | Separate trigger ranges or centralize selection |
| Timer timeout not scored | Player can become stuck at TIME: 0 | Connect timer expiration to attempt manager |
| Scene-specific controllers | Character controls differ between locations | Clear HUD prompts and transition testing |
| Inspector overrides | Documented default values may differ | Capture Inspector values before submission |
| World-object count interpretation | UI/cameras may not count as distinct world objects | Verify 30 tangible scene objects |
| Physics repeatability | Different release positions and frame timing affect score | Run multiple reproducibility tests |

## 8.5 Testing checklist

| Test | Expected result | Record |
|---|---|---|
| Locker interaction | Door opens/closes, camera returns | Pass |
| Equipment pickup | Rack javelin disappears | Pass |
| Energy drink | Stamina reaches 100%; respawns in 15s | Pass |
| Entrance light | On at ≤5m; off farther away | Pass |
| Scene entrance | Loads stadium | Pass |
| Legal throw | Projectile flight and distance result | Pass |
| Foul | 0m result and FOUL | Pass |

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|---|---|---|
| Cinemachine switch | Javelin tracked and player view restored | Pass |
| Spotlight | On at launch; off at landing | Pass |

# 10. Detailed world walkthrough and spatial design

## 10.1 Athlete preparation: player journey

The AthletePreparation scene is a compact preparatory environment rather than a second competitive arena. The athlete begins with a low stamina reserve and can navigate the locker room using third-person movement. The scene teaches spatial approach, interaction range, resource recovery, equipment collection and environmental state changes before the player transitions to the competitive space. This order creates a low-risk rehearsal of interaction controls before the timing-sensitive throw.

### Entry and orientation

The third-person camera tracks PlayerCameraRoot while WASD input moves PlayerArmature. The player is expected to identify interactive stations from visible props and context-sensitive TextMeshPro prompts. No minimap or objective arrow has been confirmed.

### Locker interaction

Approaching the locker within approximately 2.5 Unity units exposes an E-key prompt. A pivot rotates the door toward an open angle of 90 degrees, with smooth rotation set to 180 degrees per second. The Cinemachine close-up temporarily takes priority for approximately two seconds.

### Equipment collection

The javelin rack offers an E-key pickup. The displayed equipment disappears after collection. Carrying the model in the athlete's hand and enforcing equipment possession at the stadium entrance were not implemented in the confirmed build.

### Energy recovery

The energy drink station restores the PlayerStamina value to 100 percent and removes the can temporarily. Its model respawns after 15 seconds. The stamina bar starts at 25 percent and decreases by 3 units per second of horizontal movement.

### Lighting and departure

A manually operated light switch toggles two point lights. A separate proximity-based light activates within approximately five Unity units of the stadium entrance. Pressing E at the entrance loads FinalApproach_Stadium through SceneManager.

## 10.2 Stadium: competitive sequence

The stadium is a separate scene, entered through an explicit scene transition. The original Player GameObject retains PlayerRunController and ThrowController; a Starter Assets PlayerArmature child supplies the visible athlete and locomotion animation. This composition preserves the tested run-up and throw physics while improving character readability. The stadium has a runway, foul line, landing field, a score display and environmental spectator/stadium props. Exact geometry and object counts require inspection in Unity.

| Stage | Player input / condition | System response | Visible feedback |
|---|---|---|---|
| Setup | Scene loaded or next attempt begins | CameraDirector activates StadiumPlayerCamera | Athlete and throwing area framed |
| Run-up | Repeated Space presses | PlayerRunController updates movement and speed | Athlete translates; Animator Speed parameter changes |
| Preparation | Hold Left Shift | ThrowController enters preparation state | W/S angle changes are accepted |

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|---|---|---|---|
| Release | Release Left Shift | Launch velocity computed and javelin instantiated | Cinemachine follows flying javelin |
| Landing | JavelinController reports collision/landing | OnJavelinLanded and OnThrowCompleted events emitted | Distance/result and spotlight change |
| Next attempt | Enter through existing game flow | ThrowController.ResetThrow and camera reset expected | Athlete returns to setup state |

## 10.3 Spatial scale measurement worksheet

The assignment asks for concrete playable dimensions. These measurements are deliberately left as verification fields because no Unity Transform bounds or scene survey has been supplied. Measure world-space extents using the Scene view grid or the coordinates of boundary objects; state the method used.

# 11. Object-by-object design inventory

Assignment 02 requires at least 30 different objects. The following is a 30-slot inspection register, not a claim that every named candidate has been verified as a distinct implemented object. Status “confirmed” reflects the recorded development sessions

**11.01 PlayerArmature**

Location: AthletePreparation. Role and usability: Controlled avatar; collision and locomotion. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

**11.02 PlayerCameraRoot**

Location: AthletePreparation. Role and usability: Cinemachine tracking target. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

**11.03 PlayerFollowCamera**

Location: AthletePreparation. Role and usability: Third-person follow behavior. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

**11.04 Locker**

Location: AthletePreparation. Role and usability: Interactable storage prop. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

**11.05 DoorPivot**

Location: AthletePreparation. Role and usability: Rotational hinge for locker door. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object. .

**11.06 Locker door mesh**

Location: AthletePreparation. Role and usability: Visible door rotated by hinge. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

**11.07 EquipmentJavelin**

Location: AthletePreparation. Role and usability: Collectible sports implement. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

**11.08 JavelinRack**

Location: AthletePreparation. Role and usability: Equipment placement station. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

**11.09 EnergyDrinkStation**

Location: AthletePreparation. Role and usability: Consumable interaction station. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

**11.10 Energy drink can model**

Location: AthletePreparation. Role and usability: Visible collectible; respawns. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

**11.11 LightSwitch**

Location: AthletePreparation. Role and usability: Manual lighting control. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

**11.12 LockerRoomLight A**

Location: AthletePreparation. Role and usability: Toggleable room light. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.13 LockerRoomLight B

Location: AthletePreparation. Role and usability: Toggleable room light. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.14 StadiumEntrance

Location: AthletePreparation. Role and usability: Scene transition interaction. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.15 EntranceProximityLight

Location: AthletePreparation. Role and usability: Distance-triggered light. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.16 LockerRoomFloorCollider

Location: AthletePreparation. Role and usability: Solid floor boundary. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.17 InteractionCanvas

Location: AthletePreparation. Role and usability: Context prompts and stamina HUD. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object.

#### 11.18 EnergyBar

Location: AthletePreparation. Role and usability: Resource display slider. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object.

#### 11.19 LockerInteractionCamera

Location: AthletePreparation. Role and usability: Close-up Cinemachine camera. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.20 Player

Location: FinalApproach_Stadium. Role and usability: Run-up and throwing controller root. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.21 Stadium PlayerArmature

Location: FinalApproach_Stadium. Role and usability: Animated visible athlete child. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.22 StadiumPlayerCamera

Location: FinalApproach_Stadium. Role and usability: Pre-throw Cinemachine view. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.23 StadiumJavelinCamera

Location: FinalApproach_Stadium. Role and usability: Flight tracking Cinemachine view. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.24 JavelinEventSpotlight

Location: FinalApproach_Stadium. Role and usability: Launch/landing event illumination. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

#### 11.25 PreparationTimerText

Location: FinalApproach_Stadium. Role and usability: Ten-second countdown display. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

### 11.26 PreparationTimerManager

Location: FinalApproach_Stadium. Role and usability: Countdown script host. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

### 11.27 Javelin prefab instance

Location: FinalApproach_Stadium. Role and usability: Projectile with flight and landing physics. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

### 11.28 Landing field collider

Location: FinalApproach_Stadium. Role and usability: Valid landing area collision reference. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object.

### 11.29 Foul line / boundary

Location: FinalApproach_Stadium. Role and usability: Defines invalid overstep release. Evidence status: Verify visual object. Interaction classification: environmental, camera, UI or system object.

### 11.30 Scoreboard / results HUD

Location: FinalApproach_Stadium. Role and usability: Distance, attempts and outcome display. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object.

# 12. Detailed mechanics specification and compliance audit

The 20-mechanic minimum is a separate assignment requirement. The catalogue below provides a clear design description of confirmed behaviors and identifies unresolved candidates. Subsystems are described independently only when they have separate inputs, rules or outcomes. Do not present a candidate as implemented until it has been tested.

**12.01 Third-person locomotion — Confirmed**

WASD moves the preparation athlete using Starter Assets; collision constrains travel.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.02 Context-sensitive interaction range — Confirmed**

Interactable prompts appear only within a specified distance from the target.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.03 Locker door toggle — Confirmed**

E changes the locker between open and closed target rotations.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.04 Cinematic interaction focus — Confirmed**

Locker interaction temporarily raises Cinemachine close-up priority.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.05 Equipment pickup — Confirmed**

E removes the javelin model from its rack.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.06 Stamina depletion — Confirmed**

Horizontal movement reduces stamina by 3 units per second.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.07 Energy consumable recovery — Confirmed**

E restores stamina to 100 percent and removes the can.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.08 Consumable respawn — Confirmed**

The can reappears after approximately 15 seconds.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.09 Manual lighting switch — Confirmed**

E toggles two locker-room lights.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.10 Proximity lighting — Confirmed**

Entrance light automatically toggles within five units.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.11 Scene transition — Confirmed**

E at StadiumEntrance loads FinalApproach_Stadium.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.12 Rhythmic run-up — Confirmed**

Repeated Space presses drive the stadium athlete's run-up.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.13 Throw preparation — Confirmed**

Holding Left Shift enters a throw-preparation state.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.14 Angle adjustment — Confirmed**

W/S adjust the release angle while preparing; clamp 15–60 degrees.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.15 Release and projectile launch — Confirmed**

Releasing Left Shift calculates and applies javelin velocity.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.16 Foul detection — Confirmed**

A release beyond the configured throw line yields a foul.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.17 Release-quality calculation — Confirmed**

Distance from the line maps to EARLY, SAFE, GOOD, PERFECT.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.18 Speed/stability modulation — Confirmed**

Run-up speed contributes to launch velocity; stability reduces high-speed releases.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.19 Landing and scoring — Confirmed**

Landing event returns distance and quality to game result handling.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.20 Attempt reset — Confirmed**

ResetThrow clears active javelin and resets throw state.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.21 Flight camera switching — Confirmed**

OnJavelinLaunched activates the tracking Cinemachine camera.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.22 Event-driven spotlight — Confirmed**

Spotlight turns on at launch and off at landing.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

**12.23 Preparation countdown — Confirmed**

Ten-second UI countdown is confirmed

# 13. Physics and timing: precise rules

## 13.1 Release model

The supplied ThrowController defines baseThrowVelocity = 18, runContribution = 0.9, a default release angle of 35 degrees, a minimum of 15 degrees, and a maximum of 60 degrees. While preparing, W/S adjust the angle at 40 degrees per second. These are code-level values, not empirically measured real-world javelin speeds.

Distance to line = throwLineZ − player world-space Z. If this value is negative, the release is a foul and returns a failed result with distance zero. Otherwise, release quality is 0.70 when distance exceeds 3, 0.85 when above 1.5, 0.95 when above 0.5, and 1.00 at or within 0.5 units before the line.

Stability is 1.00 for player speed up to 8, 0.97 for speed greater than 8 and up to 9, and 0.93 above 9. Launch speed is (18 + playerSpeed × 0.9) × releaseQuality × stability. The projectile velocity is (0, launchSpeed × sin(angle), launchSpeed × cos(angle)). The launch is therefore directed along the positive Z axis with upward Y velocity.

## 13.2 Worked examples

| Scenario | Distance to line | Speed | Angle | Quality | Stability | Launch speed (approx.) |
|---|---|---|---|---|---|---|
| Early release | 4.0 | 7.0 | 35° | 0.70 | 1.00 | 17.01 |
| Good release | 1.0 | 8.5 | 35° | 0.95 | 0.97 | 23.64 |
| Perfect near line | 0.3 | 9.5 | 40° | 1.00 | 0.93 | 24.69 |
| Overstep foul | −0.2 | 9.0 | 35° | N/A | N/A | No projectile |

These calculations illustrate the supplied formula and are not recorded playtest scores. Actual landing distance depends on JavelinController behavior, gravity, collisions and the landing-field validation.

## 13.3 Temporal systems and state ownership

PreparationTimer starts at 10 seconds, decreases by Time.deltaTime, turns the label red in the final three seconds, and emits OnTimeExpired at zero. The user confirmed the visual countdown.

The energy-drink respawn interval is 15 seconds and the locker camera close-up is approximately two seconds. These are temporal mechanics in addition to the ten-second preparation display. Event order matters: camera and spotlight subscribe to launch/landing notifications, while the scoring system responds to OnThrowCompleted.

# 14. Prototype validation plan and evidence storyboard

## 14.1 Five action demonstrations

| Clip | Action | Before state | After state |
|---|---|---|---|
| A | Open locker | Door closed; E prompt visible | Door rotated open; close-up camera active |
| B | Collect javelin | Javelin visible on rack | Javelin disappears |
| C | Drink energy can | Stamina below maximum; can visible | Stamina full; can absent |
| D | Toggle light switch | Room lighting initial state | Two lights toggled |
| E | Throw javelin | Athlete at runway; javelin ready | Projectile launched; landing scored |

## 14.2 Cinemachine behavior evidence

Record the PlayerFollowCamera and PlayerCameraRoot in AthletePreparation; LockerInteractionCamera priority activation on E; and StadiumJavelinCamera tracking the instantiated projectile after the launch event. Capture the Main Camera Cinemachine Brain inspector and the two stadium Cinemachine cameras. The behavior distinction is follow, interaction close-up and dynamic projectile tracking—not merely three static camera objects.

## 14.3 Lighting evidence

Record manual locker-room toggle, entrance proximity toggle and stadium spotlight response to launch/landing. For each, capture an initial and activated frame from the same viewpoint. Avoid interpreting exposure changes from unrelated post-processing as proof of the intended Light component state.

## 14.4 Test cases

| Test ID | Procedure | Expected result |
|---|---|---|
| TC01 | Approach locker | Prompt appears within interaction distance |
| TC02 | Press E at locker | Door rotates and camera briefly changes |
| TC03 | Walk while stamina low | Energy bar decreases while moving |
| TC04 | Consume drink | Stamina restores and can disappears |
| TC05 | Wait 15 seconds | Can respawns |
| TC06 | Toggle switch twice | Lights change and restore |
| TC07 | Approach entrance | Proximity light activates |
| TC08 | Leave entrance | Proximity light deactivates |
| TC09 | Press E at entrance | Stadium scene loads |
| TC10 | Tap Space | Stadium athlete moves with animation |
| TC11 | Hold Shift and press W/S | Throw angle remains between 15° and 60° |
| TC12 | Release Shift | Javelin launches; camera switches |
| TC13 | Observe flight and landing | Spotlight activates then deactivates |
| TC14 | Cross foul line before release | Throw result is FOUL |